

MANSTON, United Kingdom

WMO: 037970

Lat: 51.3464N

Lon: 1.3356E

Elev: 50

StdP: 100.73

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 35047

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.0	-0.6	-5.7	2.3	-0.2	-4.2	2.7	0.4	16.5	8.9	14.8	8.8	4.2	60	0.644

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.6	25.6	18.7	23.8	18.0	22.3	17.2	19.6	24.0	18.7	22.5	17.9	21.1	5.1	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.9	13.0	21.1	17.2	12.4	20.4	16.6	11.9	19.6	55.9	24.0	53.1	22.6	50.6	21.2	22.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.0	11.4	10.2	-4.1	29.6	1.3	1.8	-5.0	30.9	-5.8	32.0	-6.6	33.0	-7.5	34.4		
			-4.7	21.5	1.4	0.7	-5.7	22.0	-6.5	22.4	-7.3	22.8	-8.3	23.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	5.1	5.3	6.9	9.3	12.2	15.2	17.6	17.8	15.4	12.1	8.3	5.8
	DBStd	5.30	2.97	2.95	2.73	2.79	2.64	2.49	2.32	2.21	2.26	2.68	2.83	3.13
	HDD10.0	654	154	134	102	46	9	0	0	0	0	11	66	133
	HDD18.3	2752	411	366	356	272	191	101	43	37	92	193	302	388
	CDD10.0	991	1	1	4	23	76	154	234	241	162	76	14	3
	CDD18.3	48	0	0	0	0	1	5	19	19	4	1	0	0
	CDH23.3	201	0	0	0	0	2	25	81	80	12	2	0	0
	CDH26.7	25	0	0	0	0	0	2	11	12	1	0	0	0
Wind	WSAvg	5.2	6.0	5.9	5.5	5.1	4.9	4.7	4.6	4.7	4.8	5.3	5.4	5.7
Precipitation	PrecAvg	575	45	34	38	42	41	46	45	58	65	66	52	
	PrecMax	878	114	124	94	128	108	104	123	177	253	250	140	105
	PrecMin	405	11	6	1	5	0	3	7	6	2	8	18	10
	PrecStd	100	22	24	23	23	21	26	26	32	48	48	33	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	13.5	15.7	20.6	23.0	26.1	28.1	28.1	25.0	20.9	15.8	13.5
		MCWB	11.2	10.5	11.7	14.4	16.7	18.9	19.6	19.4	17.8	16.9	14.3	12.2
	2%	DB	11.6	11.7	13.5	17.2	20.3	23.4	25.5	25.5	22.1	18.2	14.5	12.6
		MCWB	10.3	9.9	10.4	12.6	15.3	17.8	18.8	18.6	17.4	15.5	13.1	11.5
	5%	DB	10.6	10.6	12.2	15.1	18.4	21.4	23.7	23.5	20.4	16.9	13.6	11.7
		MCWB	9.4	8.9	9.7	11.4	14.1	16.5	18.0	18.0	16.3	14.9	12.2	10.6
	10%	DB	9.6	9.6	11.0	13.7	16.8	19.8	22.1	21.9	19.0	16.0	12.6	10.7
		MCWB	8.4	8.1	9.0	10.6	13.1	15.6	17.2	17.0	15.7	14.2	11.3	9.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.6	11.3	12.2	15.0	17.5	20.0	20.6	20.7	19.3	17.3	14.7	12.6
		MCDB	12.2	12.7	14.6	19.4	21.4	24.9	26.1	26.2	22.7	19.8	15.4	13.1
	2%	WB	10.6	10.2	11.0	13.1	15.8	18.4	19.5	19.5	18.1	16.1	13.5	11.7
		MCDB	11.4	11.4	12.8	16.2	19.3	22.3	24.2	23.6	20.9	17.6	14.3	12.4
	5%	WB	9.6	9.2	10.1	11.9	14.7	17.2	18.6	18.6	17.2	15.4	12.5	10.7
		MCDB	10.5	10.3	11.8	14.5	17.6	20.5	22.5	22.1	19.4	16.6	13.4	11.5
	10%	WB	8.6	8.4	9.2	11.0	13.6	16.1	17.7	17.8	16.4	14.6	11.4	9.7
		MCDB	9.5	9.5	10.7	13.1	16.1	19.0	21.0	20.9	18.4	15.7	12.4	10.6
Mean Daily Temperature Range	5% DB	MDBR	4.5	4.9	5.8	6.8	7.0	7.4	7.7	7.6	6.6	5.7	4.8	4.4
		MCDBR	5.3	5.8	7.4	9.0	9.7	9.9	10.2	9.8	8.3	6.5	5.3	4.7
		MCWBR	5.1	4.5	5.1	5.4	5.6	5.5	4.7	4.4	4.5	4.4	4.6	4.6
	5% WB	MCDBR	5.4	5.5	7.0	8.4	8.9	9.2	9.2	8.8	7.5	6.0	5.1	4.7
		MCWBR	5.4	4.7	5.1	5.3	5.6	5.5	4.7	4.4	4.5	4.6	4.8	4.6
Clear-Sky Solar Irradiance	taub	0.332	0.348	0.385	0.403	0.411	0.412	0.409	0.401	0.381	0.365	0.339	0.323	
	taud	2.392	2.395	2.292	2.276	2.306	2.326	2.341	2.385	2.428	2.442	2.425	2.383	
	Ebn at Noon	678	773	805	834	845	845	841	829	805	743	664	623	
	Edh at Noon	62	80	106	121	122	121	118	107	92	76	60	54	
All-Sky Solar Radiation	RadAvg	0.79	1.47	2.61	4.19	4.98	5.53	5.26	4.38	3.28	1.87	0.95	0.61	
	RadStd	0.07	0.22	0.32	0.39	0.35	0.43	0.45	0.33	0.25	0.18	0.11	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-89	N/A	N/A
(47)	Regional (73 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page